



ATLAS
EXPERIMENT



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AI-Based Energy Reconstruction for the ATLAS Tile Calorimeter at HL-LHC

Project summary and a measurement that blocks the pedestal study

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GSoC 2026 deliverables

- **Model.**

A 91-parameter quantized 1D-CNN on a 9-sample window, with a single dual-gain input $S = HG + 4096 \times LG$ and one energy output. Activations are LeakyReLU with alpha 0.25, a parameter-free bit shift.

- **Accuracy.**

The deployed quantized model reaches a calibrated per-bin band of 116.47 HG ADC and 2.97 LG ADC on the A1-A5 test set. The float student reaches 125.9 HG ADC and the 1005-parameter teacher 85.4 HG ADC on the same metric. The band is 20 bins over E_{true} in $[0, 4095]$, not a global standard deviation.

- **Firmware.**

Exported Brevitas to QONNX to hls4ml, synthesised in Vitis HLS and placed and routed in Vivado for xcku115-flva1517-2-e. One engine reaches initiation interval 1 with 200 ns latency for 366 DSP.

- **Full-chip result.**

Eleven engines cover all 77 channels of one FPGA at 280 MHz. Worst negative slack +0.012 ns, 0 of 772070 paths failing, 4026 DSP at 72.9 per cent, split 6 and 5 across the two dies by hand-written placement constraints.

- **Offline, first merge request.**

atlas/athena!90138 is merged into main, adding TileRawChannelINNMaker and TileNNEmulator to TileCalorimeter/TileRecUtils. The emulator reproduces the firmware arithmetic exactly, verified against the hls4ml C simulation with zero tolerance on integer-code equality, 4096 of 4096 simulation events and 100 of 100 testbench events.

- **Offline, second merge request.**

atlas/athena!90205 is also merged, adding the Tile.doTileNN flag next to doOpt2 and doFit. It schedules the algorithm from TileRawChannelMakerCfg, off by default and additive, writing to its own TileRawChannelINN container. Both were built and tested against release 25.0.69.

The pedestal task and why it was checked first

- The proposed method predicts the pedestal from the pulse shape. The amplitude deposited in earlier BCs leaves a long tail and the sum of those tails is the moving baseline.
- The tail is not short. Beyond the measured pulse-shape range, TilePulseShape models it as an exponential recovery.

TilePulseShape.cxx line 100

```
y = 0.00196 * (1 - exp(- x / 20664.59) ) - 0.00217;
```

- The decay constant is 20664.59 ns, about 826 BC and the asymptote is about -2.1×10^{-4} of the pulse amplitude. So the baseline at any BC depends on several hundred preceding BCs.
- Before building a correction, the question asked was whether that effect is actually present in the $\mu=200$ training sample. No job options or generation configuration were archived alongside the files, so it was measured instead.

The kernel was recovered from the sample itself

- The ROOT files store, per BC, both the ADC sample and the truth energy, for 10000 consecutive BCs and 12288 channels. Generator input and generator output are therefore both available, which makes this a linear system identification.
- In the low-gain branch the truth and the sample carry the same scale factor, so the relation has no free normalisation and the fitted form is the one below.

The model being fitted

```
sample_lo[n] = pedestal + sum_k shape(k) * ene_lo[n-k];
```

- The ADC sample was regressed on the lagged truth energies over lags -14 to +14, 40 randomly chosen channels, 398880 rows.
- Two checks confirm the fit before any conclusion is drawn from it. Neither quantity was supplied to the regression.

Self-validation

Quantity	Fitted	Expected
Pedestal	99.8598	100
In-time coefficient	0.9935	1.0

The kernel stops dead after +10 BC

- Coefficients are at the noise floor outside lags -2 to +10 and the transition is abrupt rather than gradual.

Kernel tail, coefficient by lag

Lag	Coefficient
+9	-0.002074
+10	-0.002167
+11	-0.000015
+12	+0.000043
+13	+0.000009
+14	+0.000019

- A decaying tail cannot do this. The underlying exponential has a decay constant of about 826 BC, so lag 11 should sit close to lag 10. Instead it falls by a factor of about 144 in a single step, then stays at the same magnitude as the physically meaningless negative lags.
- That is a buffer edge. TileDigitsFromPulse keeps a rolling buffer of NPulses entries, mapping to lags -10 to +10 and the default is NPulses = 21. The measured support matches that default exactly.

- **Unresolved and raised here in case it is already known.**

The measured kernel is systematically narrower than the FENICS TB2022 shape file. Measured lag +1 is 0.4848 against 0.5722 in the file, lag +2 is 0.0661 against 0.1444 and lag +3 changes sign. Summed over lags the measured kernel gives about 1.82 against 2.21. Candidate causes include a different shape file used at generation time in March and May 2025 or the per-BC pulse deformation with ImperfectionMean 1.01 and ImperfectionRms 0.02. Not proven either way.

Consequence for the pedestal study

- The simulation retains 10 BC of history and discards everything older. The accumulated baseline drift needs several hundred BC of history to build up, so it is absent from the training sample by construction.
- The proposed validation, that a predicted pedestal plus the true energy should reproduce the observed ADC count, would pass on this sample. It would confirm that a 13-tap sum can be reproduced, while claiming to validate an 826-tap effect. It would not transfer to the detector.
- The same sample also carries a constant pedestal of 100, whereas the demonstrator shows HG 148 and LG 157. Both the static offset and the drift are therefore missing from training.

- **Related earlier measurement, for context.**

On 1 demonstrator file and 1 channel, with no truth available, the network read about 33 LG ADC of energy on quiet BCs, falling to 0.88 after a crude recentring to 100.

Second open question, where the correction would run

- The method predicts the pedestal from the energies of preceding BCs. During validation those energies come from truth, which is available offline.
- In the detector there is no truth. The reconstructed amplitudes would be needed instead and those come from the network, which requires the corrected input. That is a feedback loop.
- PZC avoids this by filtering the raw ADC stream directly, which is why it fits a streaming firmware implementation with a delay of 1 BC.
- If the correction is intended to run ahead of the network in firmware, then training must emulate that same algorithm rather than subtract a truth-derived pedestal. Subtracting a truth-derived pedestal during training would produce inputs the network never sees in the detector.

Clarification requested before any implementation begins.

Decisions requested

- **Pedestal sample.**

Request regeneration of the $\mu=200$ sample with a much larger NPulses or move the study to the demonstrator streaming data and accept that it carries no truth or both.

- **Correction placement.**

Confirm whether the pedestal correction is intended to run on the raw ADC stream ahead of the network or is an offline study only.

- **Blog.**

Confirm whether the write-up goes on the HSF site or a personal site. The HSF route needs a merge request, which adds lead time.

- **Documentation.**

Confirm what should be written up and where it should live.

- **Timeline.**

The GSoC final evaluation runs 1 to 7 September. Remaining effort is planned for documentation and the write-up rather than starting new analysis.